Experiment 15
Discrete -Time Fourier Transform and Frequency -Domain Analysis

```
clc; clear; close all;
N = 10; % rectangular pulse length
n = 0:N-1;
x = ones(1,N);
w = linspace(-pi, pi, 1024); % dense frequency grid (one period)
X = zeros(size(w));
for k = 1:length(w)
    X(k) = sum(x .* exp(-1j*w(k)*n)); % direct DTFT summation
end
figure;
subplot(2,1,1); plot(w/pi, abs(X)); xlabel('\omega / \pi'); ylabel('|X(e^{j\omega})|');
title('DTFT Magnitude Spectrum'); grid on;
subplot(2,1,2); plot(w/pi, angle(X)); xlabel('\omega / \pi'); ylabel('\angle X(e^{j\omega})');
title('DTFT Phase Spectrum'); grid on;
%% Periodicity check over 2 periods
w2 = linspace(-2*pi, 2*pi, 2048);
X2 = zeros(size(w2));
for k = 1:length(w2), X2(k) = sum(x .* exp(-1j*w2(k)*n)); end
figure; plot(w2/pi, abs(X2)); xlabel('\omega / \pi'); ylabel('|X(e^{j\omega})|');
title('DTFT Magnitude over Two Periods (Periodicity = 2\pi)'); grid on;
%% Time-shift property verification
n0 = 3;
x_shift = [zeros(1,n0), x];
n_shift = 0:length(x_shift)-1;
Xs = zeros(size(w));
for k = 1:length(w), Xs(k) = sum(x_shift .* exp(-1j*w(k)*n_shift)); end
mag_err = max(abs(abs(Xs) - abs(X)));
phase_theory = -w*n0;
phase_err = max(abs(mod(angle(Xs)-angle(X)-phase_theory+pi,2*pi)-pi));
fprintf('Time-shift property: max magnitude error = %.4g\n', mag_err);
fprintf('Time-shift property: max phase-term error (rad) = %.4g\n', phase_err);
%% Linearity property verification
a = 2; b = -1;
x1 = ones(1,5); x2 = [1 0.8 0.6 0.4 0.2 0 0 0];
n1 = 0:length(x1)-1; n2 = 0:length(x2)-1;
xc = a*[x1 zeros(1,length(x2)-length(x1))] + b*x2;
nc = 0:length(xc)-1;
X1 = zeros(size(w)); X2v = zeros(size(w)); Xcomb = zeros(size(w));
for k = 1:length(w)
    X1(k) = sum(x1 .* exp(-1j*w(k)*n1));
    X2v(k) = sum(x2 .* exp(-1j*w(k)*n2));
    Xcomb(k) = sum(xc .* exp(-1j*w(k)*nc));
end
lin_err = max(abs(Xcomb - (a*X1 + b*X2v)));
fprintf('Linearity property: max complex error = %.4g\n', lin_err);
```

Code

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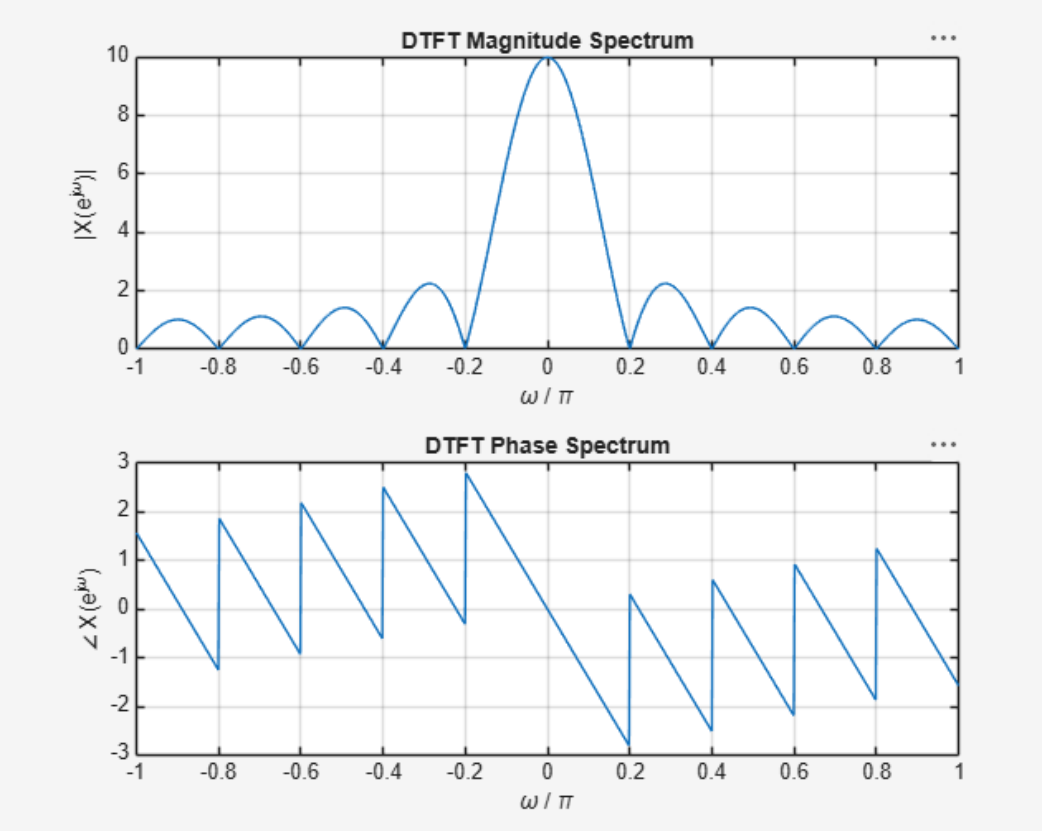


Figure 1

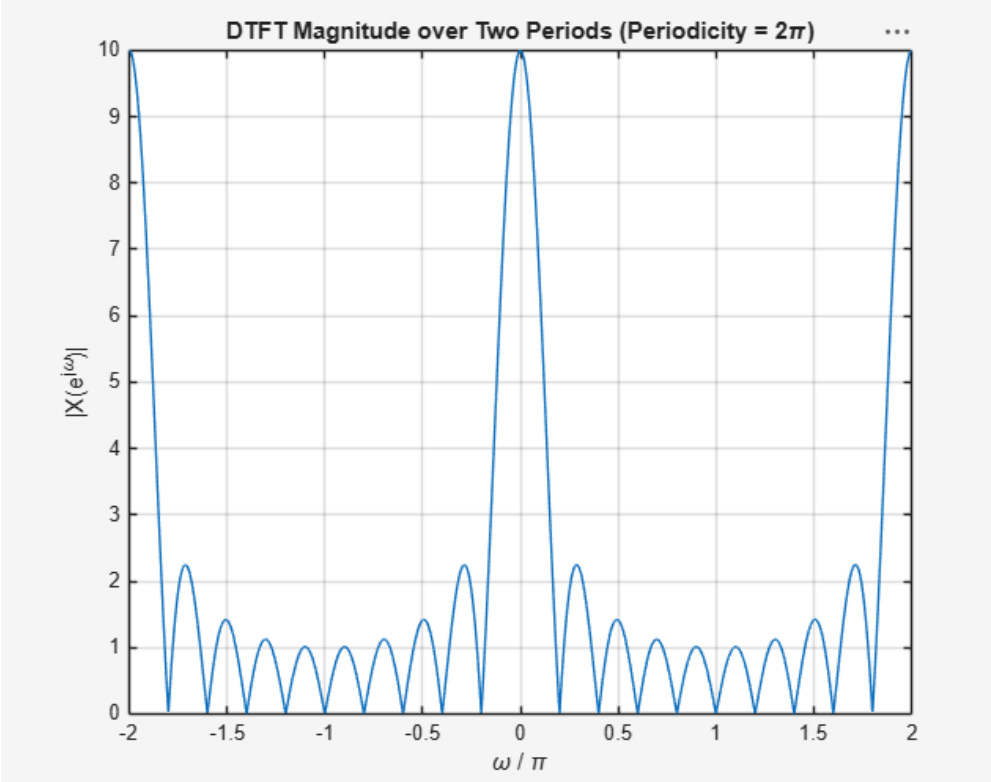


Figure 2